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RESEARCH ARTICLE

Coevolutionary Relationship Between Striatum Size and Social Play in Nonhuman Primates

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The striatum is a region of the brain specifically tied to the experience and anticipation of pleasure, reward, appropriate behavioral sequencing, cognition, learning, and social modulation. Furthermore, the striatum is connected neurologically and functionally to other brain regions associated with the exhibition of social play, such as the neocortex, cerebellum, and limbic system. For these reasons, the striatum is especially interesting to researchers of play behavior. Moreover, the caudate–putamen area of the striatum has been specifically implicated in laboratory studies of social play behavior. This study uses the phylogenetic comparative method of independent contrasts to test for an evolutionary relationship between striatum volume and a measure of social play in nonhuman primates. Relative volume of the primate striatum correlates with rate of social, but not nonsocial, play behavior across species, suggesting a coevolution of traits. The pleasurable and procedural aspects of social play behavior may be mediated in part by the striatum and further to its connection to dopaminergic pathways in the primate brain. Am. J. Primatol. 73:314–322, 2011. © 2010 Wiley-Liss, Inc.

Key words: primate; brain; evolution; striatum; social play; independent contrasts

INTRODUCTION

Historically, the study of play behavior was fraught with difficulties regarding its definition and function [e.g., Fagen, 1981]. However, animal play researchers now generally agree upon a set of five criteria that permit play to be identified in any given species [Burghardt, 2005]. Briefly, these criteria state that play behavior is (1) incompletely functional in the context in which it appears; (2) spontaneous, pleasurable, rewarding, or voluntary; (3) different from other more serious behaviors in form (e.g., exaggerated) or timing (e.g., occurring early in life before the more serious version is needed); (4) repeated, but not in abnormal and unvarying stereotypic form (e.g., rocking or pacing); and (5) initiated in the absence of severe stress [Burghardt, 2005; Graham & Burghardt, 2010].

Play represents a broad spectrum of behavioral elements including solitary locomotor–rotational play and object play, but social play has received the most attention. Social play is any playful interaction where two or more individuals engage in the behavior at the same time [e.g., Fagen, 1981]. Typically, play-fighting and rough-and-tumble play exemplify social play [Pellis & Pellis, 2009], although it could also include social interactions with objects [Tanner & Byrne, 2010]. Social play is often considered a more cognitive category of play behavior, because being successful at it requires flexible navigation of others' physical and psychological perspectives [Lewis, 2000]. As a taxonomic order, primates are generally very playful, and play has been recorded in most, if not all, primate species [Graham & Burghardt, 2010; Lewis, 2005]. Given the highly social nature of primates and the ease with which social interactions are observed, primate social play tends to dominate the primate play behavior literature.

Play is primarily associated with juveniles, although some adults in many primate species also play [Pellis & Iwaniuk, 1999a, 2000a]. However, the concentration of play during juvenility suggests that its occurrence may serve to prime the developing physiology of young mammals before reaching full adulthood [Byers & Walker, 1995; Fairbanks, 2000]. Many of these data are somewhat correlational, however, and to this end researchers interested in the precise mechanisms and adaptive functions of play have concentrated on neural, cognitive, and

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developmental aspects of the behavior [Burghardt, 2005; Pellis & Pellis, 2009]. In turn, some researchers have focused on combining this approach with phylogenetic analyses to understand a broad comparative perspective [Burghardt, 2005; Byers, 1999; Iwaniuk et al., 2001; Lewis, 2000; Lewis & Barton, 2004, 2006; Pellis & Iwaniuk, 1999a,b, 2000a,b, 2002].

It was once assumed that play was a byproduct of large brains and was most prevalent in so-called ''smart'' animals (e.g., primates and cetaceans) [e.g., Burghardt, 2005; Fagen, 1981]. Yet, Iwaniuk et al. [2001] showed that whole brain size alone does not accurately predict playfulness. Accordingly, research has since focused on different regions of the brain that may share a correlated evolutionary history with elaborations in play frequency [Burghardt, 2005; Lewis, 2000; Lewis & Barton, 2004, 2006; Pellis & Iwaniuk, 2002]. This article builds on previous research to demonstrate the importance of the striatum to this pattern of correlated evolution.

Laboratory studies on the neurobiology of play have identified several brain structures important to social play behavior, including the neocortex, cerebellum, pons, medulla oblongata, amygdala, hypothalamus, and striatum [Auger & Olesen, 2009; Burghardt, 2005; Collaer & Hines, 1995; Graham & Burghardt, 2010; Lewis, 2000, 2003; Lewis & Barton, 2004, 2006; Panksepp, 1998; Pellis & Iwaniuk, 2002; Pellis & Pellis, 2009; Pellis et al., 2010; Peters & Wozencraft, 1989; Siviy, 2010; Vanderschuren, 2010]. Damage to these areas specifically decreases social play behavior, whereas lesions to the septum, for example, increase social play behavior [Auger & Olesen, 2009; Pellis & Pellis, 2009; Vanderschuren et al., 1997]. Play is a key component of the striatal–pallidal–limbic system organization of the mammalian brain [Panksepp, 1998]. Indeed, rats with damaged caudate–putamen areas significantly reduce or cease play behavior [Panksepp, 1998]. Pellis et al. [2006] additionally note that damage to the orbital frontal cortex, part of the prefrontal cortex with connections to the limbic system, disrupts rats' ability to follow species-specific rules of play-fighting, but it does not affect their ability to play. This provides further rationale for looking at specific brain regions in attempting to understand play behavior [Pellis & Pellis, 2009; Pellis et al., 2010].

As evolutionarily ancient brain structures, the basal ganglia are especially intriguing to researchers interested in the origins of play [Burghardt, 2001, 2005]. The basal ganglia are a collection of subcortical nuclei, critically implicated with instinctual and emotional behaviors, and in learning and motor skills [e.g., Burghardt, 2001; Cardinal et al., 2002; Jueptner & Weiller, 1998]. The striatum is situated within the basal ganglia and is formed primarily of the caudate nucleus and the putamen [Ashby et al., 2010; Butler & Hodos, 1996]. Damage to this area severely and adversely affects the performance of rough-and-tumble play in rats [Panksepp, 1998], because it disrupts motor sequencing. The striatum is an important control center because it regulates a chain of neuronal responses that culminate in motor action [Kaji, 2001; Vink et al., 2005]. It both receives input from and also projects to many regions of the cortex, including the frontal and parietal cortices, thalamus, and other areas of the basal ganglia (i.e., globus pallidus and substantia nigra) to form basal ganglia–thalamo–cortical loops [Alexander & Crutcher, 1990; Ashby et al., 2010; Jueptner & Weiller, 1998; Rolls, 1994; Vink et al., 2005]. These loops are important for selecting movements and appropriate muscle activation to perform motor tasks [Jueptner & Weiller, 1998]. In other words, the striatum is crucially involved in preparing and executing motor patterns [Alexander & Crutcher, 1990; Hikosaka, 1998; Mink, 1996; Pellis & Pellis, 2009; Vink et al., 2005]. Furthermore, striatal projections to the cerebellum work to monitor and optimize appropriate motor decisions [Jueptner & Weiller, 1998]. These neural systems are implicated in flexible use of motor actions, which are crucial to the execution of play behavior.

The striatum and other basal ganglia are implicated in learning and motor skills most critically during development, when individuals gain experience. Juvenile experiences in play, including different amounts, types, and partners, certainly influence physiological and neural development [Byers & Walker, 1995; Pellis & Pellis, 2009]. Rats deprived of play as juveniles may subsequently be unable to engage in a normal range of socially mediated behaviors as adults, owing to changes in neural circuitry during development [Pellis & Pellis, 2009; Pellis et al., 2010]. For example, more playful individuals tend to have more thickly myelinated brains and thicker cortices [Lewis, 2010; Pellis & Pellis, 2009; Pellis et al., 1992]. Moreover, rats reared with nonplayful partners have abnormal development of their striatal neurons [Pellis et al., 1992]. Pellis et al. [1993] specifically identified that the striatum efficiently organizes the selection of defensive tactics to be used in play. Furthermore, in primates, it is the putamen that receives the primary projections from motor cortical areas that control muscle movements in the limbs and face [Alexander & Crutcher, 1990]. Careful selection of motor actions during social play is crucial for the success of a play bout. Together with evidence suggesting the importance of the cerebellum to social play [Lewis & Barton, 2004] and given striatal projections to the cerebellum [Jueptner & Weiller, 1998], these findings suggest the importance of both motor and cognitive proficiency to the exhibition of social play in mammals.

Dopamine, a neurotransmitter, is key to the experience of pleasure; and dopamine levels increase

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in response to physical activity [Auger & Olesen, 2009; Lewis, 2010; Pellis & Pellis, 2009]. The role of dopamine in the basal ganglia is complex, but dopaminergic inputs seem to stimulate striatal neurons [Alexander & Crutcher, 1990], and it is likely that dopamine in the striatum initiates mechanisms that regulate striatal synapses [Tang et al., 2001]. Such effects may reinforce cortical action in other parts of the basal ganglia [Alexander & Crutcher, 1990]. In lizards, the basal ganglia assist learning [Punzo, 1985], and in mammals they also contribute to play-fighting [Burghardt, 2005; Panksepp, 1998; Siviy, 1998]. The striatum factors into human play behavior during the acquisition of skills in video games and may be directly involved with gaming addiction [Erikson et al., 2010; Graham, 2010]. Because striatal structures are intimately connected to dopaminergic pathways, and thus to the experience and anticipation of pleasure and reward, as well as to decision making, goal directedness, habit formation, learning, sequential behavior, motor control, cognitive flexibility, and social modulation [Ashby et al., 2010; Atallah et al., 2007; Burghardt, 2005; Butler & Hodos, 1996; Elliott et al., 2003; Erikson et al., 2010; Tang et al., 2001; Vink et al., 2005], the relationship between play behavior and the striatum is especially intriguing.

METHODS

The research presented here used published literature and was, therefore, exempt from Institutional Animal Care Committee review. This research adhered to the legal requirements of the country in which it was conducted and to those of the American Society of Primatologists Principles for the Ethical Treatment of Nonhuman Primates.

The primate behavior literature (including some theses, dissertations, and occasionally unpublished data) was searched for time budget and frequency measures of social play. Only reports that explicitly used the term ''social play'' were included. Time budget or frequency measures of social play were preferentially sought. Time budgets provide a percentage of daily activity investment for each behavioral category, and thus broadly represent a practicable way to measure investment in significant behaviors across groups or species [Post & Baulu, 1978]. Play behavior forms a part of an animal's daily time budget, and so the percentage of time spent in social and nonsocial play are independent of one another and do not sum to 100%. Elsewhere, where sufficient and appropriate information was available, raw data were transformed through simple calculations to give a mean time budget percentage for social play in each species, based on the length of the study. For example, on the basis of 170 hr of observation, King et al. [1980] reported the number of minutes per hour spent engaged in social play for each of seven chimpanzees (*Pan troglodytes*); the mean of these scores was calculated as 5.3%. Where two or more suitable species replicates were found in the literature, a time budget percentage of social play was taken for each study and a mean calculated to give a single value for each species. For example, three studies on gorillas (*Gorilla gorilla*) stated time budget means for social play as 22.2% [Fossey, 1979], 6.4% [Fischer & Nadler, 1978], and 15.7% [Freeman & Alcock, 1973]. These totals were summed and divided by the number of studies to give a species average: $44.3/3 = 14.8$. This value was log transformed before phylogenetic analysis to meet the assumptions of the test [Lewis & Barton, 2004, 2006; Martins & Hansen, 1997; Nunn & Barton, 2001]. Nonsocial play measures followed the same criteria, although authors rarely use this term explicitly. In this study, the term ''nonsocial play'' refers to any play that was not associated with a partner or dyad (i.e., solitary play and nonsocial object play). Given the criteria used for data collation, it is possible that the data are not exhaustive but they likely remain representative, both of the available literature and of the species. Data from wild and captive studies were combined to increase the sample size. Although rates of play tend to be higher in captive studies owing to the removal of predation and foraging pressures in captivity, captive data may demonstrate the species' potential for engaging in behaviors, such as social play. In addition to data on play behavior, adult brain and body weights were taken from Barton [1999] and striatum volumes were taken from Stephan et al. [1981]. Striatum size was subtracted from brain size before regression, and the confounding effects of body size and whole brain size were removed by multiple regressions in order to test for associations specific to the striatum [e.g., Jerison, 1973].

The method of independent contrasts was used to analyze these data. Independent contrasts provide a technique for reconstructing evolutionary change from comparative data on extant species. It assumes that each change on a given branch of a phylogeny occurs independently from changes in other parts of the tree. Such independence of data points permits the use of standard statistical tests. If changes in y are correlated with changes in x, it suggests a coevolution of those traits. For this reason, one primary use of independent contrasts is to test hypotheses concerning evolutionary change [e.g., Felsenstein, 1985]. The necessary primate phylogeny, inclusive of branch lengths, was taken from Purvis [1995]. The phylogenetic analyses were performed using the Comparative Analysis of Independent Contrasts [CAIC] software package, using the CRUNCH algorithm for continuous variables [Purvis & Rambaut, 1995]. Unequal branch lengths were used based on Purvis [1995] and assumptions were checked using the diagnostics

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suggested in the CAIC manual. Statistical analyses were performed on the output using StatView v. 5.0. All data were log transformed before analysis [Freckleton, 2000] and regressions were set through the origin [Purvis & Rambaut, 1995]. The above methods and data have also been described in detail elsewhere [Lewis, 2003; Lewis & Barton, 2004, 2006].

RESULTS

Table I details the social play, body size, brain size, and striatum volumes for 14 primate species.

The correlation between social play and striatum size was positive and significant ($F = 4.93$, df = 12, $b = 1.43$, $r^2 = 0.41$, $P < 0.01$) (see Fig. 1). Striatum size was not, however, correlated with nonsocial play ($F = 1.04$, df = 9, $b = 1.94$, $r^2 = 0.17$, $P = 0.35$).

The results show that relative volume of the primate striatum correlates with the percentage of social, but not nonsocial, play behavior across primates, suggesting a coevolution of traits.

DISCUSSION

The striatum has been previously implicated in social play [e.g., Panksepp, 1998; Pellis & Pellis,

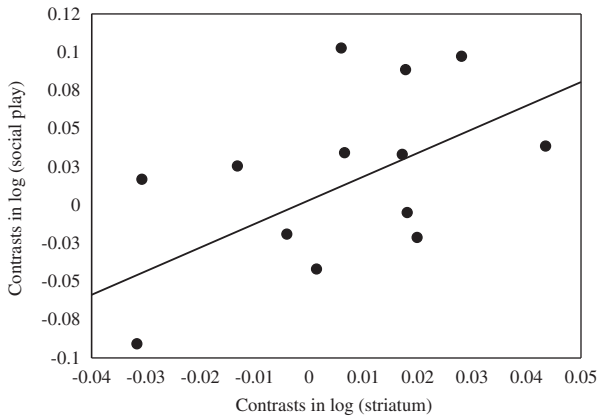


Fig. 1. Contrasts in log (social play) on log (striatum) in primates.

TABLE I. Percentages of Social and Nonsocial Play, Adult Body and Brain Weights and Striatum Volumes in Nonhuman Primates

Species	% social play[a]	% nonsocial play[a]	Brain size (g)[b]	Body size (g)[c]	Striatum (mm^3)[d]	Sources
Callithrix jacchus	1.32	0.89	7.90	288	372	Lewis, 1997[e]; Stevenson and Poole, 1976, 1982[e]; Voland, 1977[e]
Cebus albifrons	6.17	3.18	82.0	2,489	3,258	Bernstein, 1965[e]; Marshall, 1998[e]
Cercocebus albigena	9.00	–	99.1	7,362	4,146	Cahill, 1978[e]
Cercopithecus mitis	4.00	–	75.0	5,821	2,733	Lawes and Piper, 1992[f]
Colobus badius	3.38	2.41	73.8	7,998	3,217	Clutton-Brock, 1974[f]; Marsh, 1981[f]; Struhsaker, 1975[f]
Erythrocebus patas	7.81	–	106.6	8,690	3,624	Rowell and Chism, 1986[e]
Gorilla gorilla	14.79	2.55	505.9	114,551	14,567	Fischer and Nadler, 1978[e]; Fossey, 1979[f]; Freeman and Alcock, 1973[e]
Hylobates lar	12.30	0.50	107.7	5,559	4,784	Bartlett, 1999[f]; Bernstein and Schusterman, 1964[e]
Macaca mulatta	5.50	1.28	95.10	5,902	4,032	Chamove, 1973[e]; Ehardt and Bernstein, 1987[e]; Gard and Meier, 1977[e]; Post and Baulu, 1978[e]
Nycticebus coucang	1.01	1.20	10.0	659	760	Glassman and Wells, 1984[e]; Lewis, 1997[e]
Pan troglodytes	22.49	8.97	410.3	37,844	12,246	Bloomsmith, 1989[e]; Bloomsmith et al., 1994[e]; King et al., 1980[e]; Kraemer, 1979[f]; Kraemer et al., 1982[f]; Lewis, 2000[e]; Mendoza-Granados and Sommer, 1995[e]
Papio spp.	12.50	5.51	172.8	17,623	7,182	Chalmers, 1980[f]; Cheney, 1978[f]; Coelho and Bramblett, 1982[e]; Owens, 1975a,b[f]; Rose, 1977[f]; Young and Hankins, 1979[e]; Young et al., 1982[e]
Pithecia spp.	2.98	4.20	34.9	2,019	1,918	Dugmore, 1986[e]; Potter, 1999[e]
Saguinus oedipus	1.80	1.87	10.0	417	453	Potter, 1999[e]

[a]Social and nonsocial play measures are mean percentages derived from the sources provided in this table.
[b]Mean adult brain size from Barton [1999].
[c]Mean adult body size from Barton [1999].
[d]Striatum volume from Stephan et al. [1981].
[e]Captive study.
[f]Wild study.

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2009; Pellis et al., 1993], but this is the first study to use independent contrasts to demonstrate an evolutionary relationship between striatum size and social play behavior in nonhuman primates. That this pattern holds for social but not for nonsocial play, and because it is found after controlling for body and brain size, suggests that the striatum is a neural component specifically associated with the performance of social play in primates.

This result supports previous studies that show an association between play and the striatum. For example, a similar finding by Lewis [2003] [Burghardt, 2005] showed that striatum volume coevolved with total play frequency in primates. Given the findings of this study, inclusion of social play within the category of total play likely contributed significantly to the Lewis [2003] results. More specifically, Panksepp [1998] demonstrated that caudate–putamen damage in rats significantly reduced rough-and-tumble play. Pellis et al. [1993] showed that injecting dopamine inhibitor drugs into the brains of juvenile rats disrupts normal dopamine function in the striatum. The effect is that while the rats still played, it diminished the cohesiveness of their social play because they were unable to choose appropriate motor sequences. Thus, species differences in offensive and defensive tactics during rough-and-tumble play are likely owing to differences in striatum circuitry [Pellis & Pellis, 2009]. These findings suggest that both motor and cognitive skills are important to the exhibition of social play in mammals, and perhaps that the striatum functions to select the best action patterns in successful social play bouts [Hikosaka, 1998; Mink, 1996; Pellis & Pellis, 2009].

The striatum seems to be part of a system of neural regions, specifically tied to the exhibition and performance of social play behavior and forms a functional network with other parts of the brain, such as the neocortex, cerebellum, and parts of the limbic system that are also implicated in social play. The projections and circuitry between the striatum and these other neural regions influence motor and cognitive function, including planning action, learning, and procedural memory [Bostan et al., 2010; Burghardt, 2001, 2005; Diamond, 2000; Erikson et al., 2010; Graybiel, 2008; Jog et al., 1999; Lewis, 2000; Lewis & Barton, 2004, 2006; Pellis & Iwaniuk, 2002; Pellis & Pellis, 2009]. These visual and motor elements are crucial to the expression of play, where reliance on vision and the innate understanding of motor actions is vital.

Vision is the dominant sensory domain of primates [Barton, 1998], and visual elements, such as facial expressions, hold increased relevance in primate social cognition and social play behavior [Barton, 1998; Dobson, 2009; Lewis, 2003]. Social play is considered an especially cognitive form of play behavior because it relies not only on physical movement, but also on reading (social) cues appropriately [e.g., Lewis & Barton, 2004]. Primates use a rich and complex variety of facial expressions, and it is likely that neural visual and facial motor control areas (e.g., V1, facial nuclei) coevolved with social group size to facilitate social interactions [Dobson & Sherwood, 2010]. Control over facial muscles is, thus, imperative in primate sociocognition; this would include social play, where a play face (a relaxed, open-mouthed expression exhibited during play behavior) is a key social signal indicating playful, rather than aggressive, intent [Bekoff, 1975; Dobson, 2009; Palagi, 2006; Pellis & Pellis, 1996]. That it is the primate putamen that receives primary projections from motor cortical areas controlling facial muscle movements [Alexander & Crutcher, 1990] further suggests the importance of the striatum to the exhibition of social play behavior.

In monkeys, the striatum is involved in both initiating and suppressing movement [Lebedev & Nelson, 1999; Raemaekers et al., 2002]. Vink et al. [2005] found that in humans, neuronal activity in the striatum increased when subjects tried to hold back on planned movement, suggesting striatal involvement in inhibitory motor control. Such abilities are critical to the performance of social play behavior, for example, in self-handicapping, where older or stronger partners suppress their strength and experience to participate successfully in social play with smaller or younger partners. Although juvenile primates generally prefer play partners of their own age and sex, sometimes potential play partner availability is limited to nonmatched peers [Fagen, 1981; Lewis, 2005]. In such cases, self-handicapping is essential for the continuance of play. This reflects a certain amount of cooperation or social cognitive competence [Bekoff & Allen, 1998; Smith et al., 1999; Sussman et al., 2005; Watson & Croft, 1993], and may have evolved to ensure appropriate motor functioning during development—skills that can be honed during juvenile play.

Experiencing reward is crucial to risk-taking behavior [Grabenhorst et al., 2010; Izuma et al., 2008], which may include some patterns of social play. The exhilaration of engaging in risky behavior and successfully executing new skills is owing to dopamine uptake in the striatum and other parts of the brain [Ashby et al., 2010; Schultz, 2002]. Young animals exhibit exuberance in their playful interactions and seem compelled to take both physical and social risks. Yet, risk taking is not necessarily maladaptive, if doing so helps to learn and improve new physical and social skills, which might ultimately benefit the player during urgent or unpredictable situations [Spinka et al., 2001]. For example, misjudging a branch during playful chase presents a real danger, but learning physical maneuvers to still reach safety might eventually prove invaluable (e.g., predation, broken branches). As such, regions of the

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limbic system (e.g., amygdala, hypothalamus) and basal ganglia, including the striatum (specifically the caudate–putamen area), have been identified as important structures in the expression of play behavior [Burghardt, 2001, 2005; Lewis & Barton, 2006; Panksepp, 1998; Pellis & Pellis, 2009; Pellis et al., 1993].

Among human adolescents, the propensity to take risks may be owing to developmental differences in dopamine signaling [Cohen et al., 2010]. Cohen et al. [2010] demonstrated that adolescent reward-seeking is especially sensitive owing to heightened dopaminergic prediction error responsivity in the striatum. Indeed, the striatum is implicated in response to both actual reward and to the anticipation of predicted rewards [Elliott et al., 2003]. Experience helps to fine-tune the ability to predict reward, but because adolescents lack sufficient experience to exert control over reward-driven urges, they are more prone to taking risks. This may explain why adolescents take more risks; but risks taken during play could be adaptive if performing the behavior provides new skills or confidence during an emergency [Špinka et al., 2001].

Striatal control over learning, risk taking, motor planning, and cognition, and its connection to other neural areas, all point to its possible involvement in play complexity. Owing to limitations of the available literature, this study focused on time budgets of play, with the assumption that increased time investment reflects a measure of how important the behavior is to an individual or species. Although it is possible that time spent playing is a proxy measure for play complexity, it would be interesting to test for a more direct measure of play complexity. Although social play is widely considered more cognitively challenging than nonsocial play, complex motor actions need not be entirely limited to social play; risk-taking and cognitive challenges may also occur during nonsocial play sequences. Future studies should test whether or not the relationship between striatum size and play holds with play complexity measures, and if this increases or decreases the correlation. Devising behavioral complexity indices is challenging [Burghardt, 2005; Byers, 1999; Reader & Laland, 2002], but may be an especially fruitful area of research for a diverse behavioral category, such as play behavior.

This study demonstrates an evolutionary relationship between social play and striatum size in nonhuman primates. The striatum is a neural region implicated in various motor and cognitive functions [Jog et al., 1999; Jueptner & Weiller, 1998] that may ensure the appropriate and successful exhibition of play between partners. The basal ganglia are ancient structures, so perhaps both motor and cognitive functions have ancient coevolutionary relationships that work together to drive behaviors like social play, which require both cognitive and motor capabilities. It is likely that play behavior arose more than once across broad taxonomic groups in response to proximate processes, such as instinctive behavior [Burghardt, 2005]. This may go some way to explaining why play behavior seems intimately connected to various different behavior systems, borrowing sequences from other behaviors typically considered more serious (e.g., sex, aggression, etc.). Alternatively, this may be a slight adaptation from the ancestral form [Burghardt, 2001; Katz & Harris-Warrick, 1999]. The striatum is connected neurologically and functionally to other brain regions that are also crucial to the exhibition of social play, such as the neocortex, cerebellum, and amygdala [e.g., Burghardt, 2005; Jueptner & Weiller, 1998; Lewis, 2000; Lewis & Barton, 2004, 2006; Pellis & Iwaniuk, 2002; Pellis & Pellis, 2009], and further to dopaminergic inputs connected to pleasurable experiences [Ashby et al., 2010; Schultz, 2002]. That the striatum fits into a similar phylogenetic relationship with social play is compelling if not entirely unsurprising, given its role in motor control, sequencing and action, learning, pleasure, and reward. As such, the striatum seems to be part of a crucial functional neural system that assists motor and cognitive control used in social play behavior, at least in nonhuman primates.

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